

Problem ID	Problem Statement	Brief Description	Expected ML Model(s) / Technique(s)	Dataset Clues (examples only)
P1	Employee Attrition Prediction & Workforce Risk Analysis (IT / BFSI / Healthcare / Manufacturing / Retail, etc.)	Organizations face significant challenges when experienced employees leave the company, affecting productivity, operational continuity, and recruitment costs. The objective of this project is to build a supervised machine learning model to predict whether an employee is likely to leave the organization based on demographic, performance, engagement, compensation, and work-related attributes. Students are expected to perform data preprocessing, exploratory data analysis, feature engineering, model training, hyperparameter tuning, model evaluation, feature importance analysis, and comparison of multiple supervised learning algorithms. The project should focus on improving model generalization performance, reducing overfitting, and analysing the impact of different hyperparameter settings on model behavior.	Logistic Regression / Decision Tree / Random Forest / Gradient Boosting / KNN	IBM HR Attrition Dataset (Kaggle)/ Employee Attrition datasets from UCI/Kaggle / etc.
P2	Retail Sales Amount Prediction & Customer Spending Analysis	Retail and e-commerce organizations aim to understand the factors influencing customer spending patterns to improve promotions, inventory planning, and customer engagement strategies. The objective of this project is to build a supervised machine learning regression model to predict customer purchase amount or sales value using transactional, demographic, seasonal, and behavioral features. Students are expected to perform preprocessing, EDA, feature engineering, regression model training, hyperparameter tuning, model evaluation, residual analysis, and comparison of multiple regression algorithms.	Linear Regression / Decision Tree Regressor / Random Forest Regressor / Gradient Boosting Regressor / KNN Regressor	Supermarket Sales Dataset (Kaggle)/ BigMart Sales Dataset (Kaggle)/ Retail Sales datasets from UCI/Kaggle / etc.

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		The project should emphasize model refinement, feature impact analysis, and performance improvement through tuning and validation strategies.		
P3	Customer Behaviour Segmentation & Engagement Profiling (Retail / Banking / Telecom / Subscription Services, etc.)	Organizations use customer segmentation to understand different behavioral patterns and to design targeted marketing, retention, and engagement strategies. The objective of this project is to apply unsupervised learning techniques to identify meaningful customer segments using transactional, demographic, engagement, and usage-related attributes. Students are expected to perform data preprocessing, exploratory analysis, feature engineering, clustering model building, cluster validation, hyperparameter experimentation (e.g., number of clusters, distance metrics, DBSCAN parameters), and interpretation of cluster characteristics. The project should focus on analysing cluster quality, stability, interpretability, and business relevance.	K-Means / Hierarchical Clustering / DBSCAN	Customer Personality Analysis Dataset (Kaggle)/ Mall Customer Segmentation Dataset (Kaggle)/ Online Retail Dataset (UCI)/ etc.
P4	ANN-Based Customer Response Prediction (Banking / Telecom / Retail / E-Commerce / EdTech, etc.)	Organizations across various domains run campaigns to improve customer engagement, subscription renewal, product adoption, or offer acceptance. The objective of this project is to build an Artificial Neural Network (ANN)-based model to predict whether a customer is likely to respond positively to a campaign or offer using demographic, transactional, and behavioral features. Students are expected to perform required data preprocessing, exploratory data analysis, feature engineering, ANN model construction, backpropagation-based training, optimizer experimentation, hyperparameter	Perceptron / Multi-Layer Perceptron (MLP) / Feed Forward ANN/etc.	Bank Marketing Dataset (UCI)/ Marketing Campaign Dataset (Kaggle)/ Customer Personality Analysis Dataset (Kaggle)/ etc.

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		tuning, regularization, model evaluation, and ANN diagnostics. The project expects emphasize ANN refinement, convergence behavior analysis, reduction of overfitting, and improvement of model generalization performance.		
P5	ANN-Based Product Quality Classification (Manufacturing / Industrial Systems / Electronics, etc.)	Manufacturing and industrial systems continuously monitor operational and process-related parameters to maintain product quality and reduce defects. The objective of this project is to build an ANN-based classification model to predict product quality category or defect occurrence using structured operational and process-related features. Students are expected to perform data preprocessing, exploratory data analysis, feature engineering, ANN architecture design, backpropagation-based training, optimizer comparison, regularization, hyperparameter tuning, training diagnostics, and model evaluation. The project should focus on analysing ANN learning behavior, improving convergence stability, reducing overfitting, and studying the impact of various ANN training configurations on model performance.	Multi-Layer Perceptron (MLP) / Feed Forward ANN/etc.	Steel Plate Fault Dataset (UCI)/ Sensorless Drive Diagnosis Dataset (UCI)/ Manufacturing Quality datasets from UCI/Kaggle / etc.